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# How ChatGPT impacts student engagement from a systematic review and meta-analysis study

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ABSTRACT

Generative artificial intelligence, such as ChatGPT, has been increasingly integrated into education to change student learning experience. Current empirical studies have mixed results on how ChatGPT impacts student behavioral, cognitive, and emotional engagement. This systematic literature review and meta-analysis explores whether and how ChatGPT impacts student behavioral, cognitive, and emotional engagement. We used the PRISMA method to select, analyze, and report the results. We screened 766 articles from four databases and identified 17 empirical studies with 1735 students for analysis. We compared the effect on student engagement between ChatGPT-based and non-ChatGPT learning. We found a medium effect size on overall student engagement in ChatGPT-based learning in the random effects model. Our analyses further suggest that ChatGPT-based learning is more effective in fostering student behavioral (medium effective size), cognitive (large effective size), and emotional engagement (medium effective size) than non-ChatGPT learning. Our findings revealed ChatGPT is an effective tool for engaging students in learning. We also suggested three roles ChatGPT plays in fostering student engagement: personalized tutoring, programming and technical assistance, and content generation and collaboration. Our systematic literature review revealed potential risks and results in student disengagement, such as over-reliance.

## 1. Introduction

Generative artificial intelligence (GenAI) technology has the potential to generate educational content, improve student engagement and interaction, and provide personalized learning experiences, given its capability to generate texts, images, and videos (Chiu, 2023; Kasneci et al., 2023). ChatGPT, a popular GenAI technology, has been integrated into a range of tasks for diverse disciplines, such as translation and essay writing for language learning and computational thinking tasks for computer education, thereby supporting the development of literacy and problem-solving skills (Bilal Khallel, 2024; Kasneci et al., 2023). Its impact on student engagement is multifaceted, with evidence suggesting both positive and negative effects. Student engagement is viewed as multidimensional, including behavioral, cognitive, and emotional components (Chiu, 2021, 2022; Christenson et al., 2012; Fredricks et al., 2004; Sinatra et al., 2015), which is associated with student learning attention, academic performance, and emotional wellbeing. Behavioral engagement refers to student attention, participation, effort, intensity, or persistence in learning activities. Cognitive engagement refers to how much mental effort students put into applying complex learning strategies to complete learning assignments. Emotional engagement includes student sentiments about teachers, peers, learning activities, school, and belonging (Christenson et al., 2012; Fredricks et al., 2004; Sinatra et al., 2015). Moreover, student engagement is a prerequisite for effective learning and requires intrinsic motivation (Reeve, 2013), and more student learning activities with GenAI such as ChatGPT will become more common (Chiu, 2024). Understanding whether and how ChatGPT affects student engagement adds to the academic and practitioner community for AI in education. This understanding benefits our student learning in future AI-powered classrooms.

Literature suggests ChatGPT can enhance student motivation and engagement in learning processes by providing immediate feedback and creating a more interactive learning environment (Kasneci et al., 2023). ChatGPT offers a low-stress learning environment to practice language

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skills and engage with content, which can help build confidence and fluency (Escalante et al., 2023; Pellas, 2023). This supportive environment may encourage students to participate more actively in their learning. However, ChatGPT may also raise concerns about disengagement. Students may be inclined to rely on AI for academic work, potentially undermining their learning and integrity (Dindorf et al., 2024). This reliance on ChatGPT may result in reduced critical thinking abilities. Students may become overly reliant on AI-generated materials, hindering their capacity to connect deeply with the material and acquire important analytical abilities (Al Shloul et al., 2024). These may suggest that the behavioral and cognitive aspects of engagement need more investigation. Moreover, students' perceptions of using ChatGPT in their learning are mixed. Some students report increased satisfaction and interest (Alneyadi & Wardat, 2023; Escalante et al., 2023), while others experience disappointment or anxiety when they rely too heavily on the tool (Stojanov et al., 2024). This conflict shows that the emotional aspects of engagement require additional investigation. Overall, ChatGPT has the potential to engage and disengage students in learning. It is necessary to have a synthesis study to help educators and researchers navigate these complexities for integrating AI thoughtfully into education.

This study aims to investigate how ChatGPT impacts student engagement. The findings are expected to ensure ChatGPT serves as a tool for enhancement rather than a crutch for disengagement. The findings may encourage teachers to develop strategies that leverage the strengths of ChatGPT while mitigating its drawbacks in student engagement.

## 2. Literature review

### 2.1. ChatGPT in education

Chatbots as text-based communication agents were first developed in the early 2000s for language learning, but they tended to provide vague and inaccurate responses that sometimes mislead students (Fryer et al., 2019; Yin et al., 2021). Therefore, students had relatively limited engagement with the chatbots due to their limited educational values. With the advance of GenAI, an AI-empowered chatbot with enhanced conversational capability built on a GPT-3.5 large language model, ChatGPT, was launched to support natural language processing that allows humans to interact with it in conversational language (Chiu, 2023, 2024). Unlike old-generation rule-based chatbots, which can only provide predefined and fixed responses based on students' responses, ChatGPT can flexibly handle more complex and adaptable contexts (Ng et al., 2024). Relevant studies highlighted the potential of ChatGPT in supporting group interactions in the online learning environment by facilitating group discussions and providing explanations on certain topics (Bilal Khallel, 2024; Javaid et al., 2023). It handles interactions with multiple users spontaneously. Some educators applied ChatGPT as a personalized feedback tool to guide students through the learning activities, while some educators treated ChatGPT as an intelligent tutoring system to provide relevant information and materials to promote adaptive learning and self-regulated learning. Therefore, ChatGPT could impact student engagement.

### 2.2. Student engagement and ChatGPT

Student engagement has been studied by researchers to understand the relationship between coursework and student learning (Axelson & Flick, 2010), the amount of student effort and knowledge, and the impacts from environmental and individual factors (Bakker et al., 2015). Student engagement refers to students' active involvement in the learning activities and commitment to fulfilling the learning outcomes (Christenson et al., 2012). It can be described in a multifaceted construct (Sinatra et al., 2015): Behavioral, emotional and cognitive engagement. Behavioral engagement related to students' participation, attention, and effort; emotional engagement associated with task-facilitating emotions and reactions; cognitive engagement, which refers to the degree of investment in learning and mastering the content (Christenson et al., 2012; Fredricks et al., 2004; Kahu, 2013; Sinatra et al., 2015). Indicators of behavioral engagement include the duration and effort spent in learning activity participation and interaction with others (Fredricks et al., 2004; Kahu, 2013; Lam et al., 2012; Sinatra et al., 2015). Indicators of emotional engagement include attitudes, interests, and values towards learning, while indicators of cognitive engagement include students' motivation to learn, self-regulation, active construction of knowledge, deep processing of information, and persistence to achieve academic requirements (Fredricks et al., 2004; Kahu, 2013). Additionally, researchers organized the engagement process into three areas: factors influencing student engagement (such as teaching practices), indicators of student engagement (such as interaction with students and instructors), and outcomes of student engagement (such as academic achievement) (Kahu, 2013; Lam et al., 2012). This review examines the literature to explore how ChatGPT impacts student engagement based on the above indicators in each construct (Kahu, 2013; Lam et al., 2012).

ChatGPT has the potential to foster behavioral, cognitive, and emotional engagement through adaptive learning. Adaptive learning is defined as delivering learning materials determined by students' interaction in a previous stage (Kerr, 2016). This form of learning incorporates computer algorithms and artificial intelligence to provide personalized learning materials and experience, transforming students' roles from passive receivers to active collaborators through a scaffold approach (Deng & Yu, 2023). ChatGPT serves as a virtual agent to engage students in asking questions through prompting. In individual learning, this large language model generates questions and encourages students to think critically in their learning activities (Kasneci et al., 2023). In group and remote learning, ChatGPT facilitates group discussions and debates by providing a discussion structure, real-time feedback, and personalized guidance to students, where they interact with one another and the chatbot, sustaining their attention in the learning activities. This form of guidance improves student behavioral engagement through participation. When students actively participate in the learning activities with ChatGPT, they construct knowledge in a ChatGPT-mediated supportive learning environment through a constructivist approach (Boghossian, 2006) by generating questions that stimulate students' interests and providing real-time constructive and personalized feedback. The impact on students' interests could impact their emotional and cognitive engagement, including their learning motivation, attitudes, and values toward using ChatGPT and other emerging technologies (Huesca et al., 2024). Regarding cognitive engagement, self-regulation, one of the indicators, refers to learners' ability to control, monitor, and regulate their learning process independently in attaining learning goals, emphasizing three phases, namely forethought, performance, and self-reflection (Zimmerman, 2000, 2008). As Zimmerman (2000) mentioned, self-regulation can be affected by personal, behavioral and environmental factors, involving observing, monitoring, and adjusting performance processes, conditions, and affective states in a cyclical cycle. Strategic use of ChatGPT resolves students' difficulties by providing useful information in the performance phase, therefore promoting their active construction of knowledge and deep processing of information, preventing students from being interrupted by learning difficulties and strengthening their persistence to achieve learning goals (Wu et al., 2024). ChatGPT can also record past questions and answers during the self-reflection phase, encouraging students to reflect critically on their strengths and weaknesses in the concepts. This can support the development of self-efficacy and contribute to their academic achievements.

In sum, as discussed in this section, we use the indicators shown in Fig. 1 to map student engagement into behavioral, emotional, and cognitive dimensions for this study.

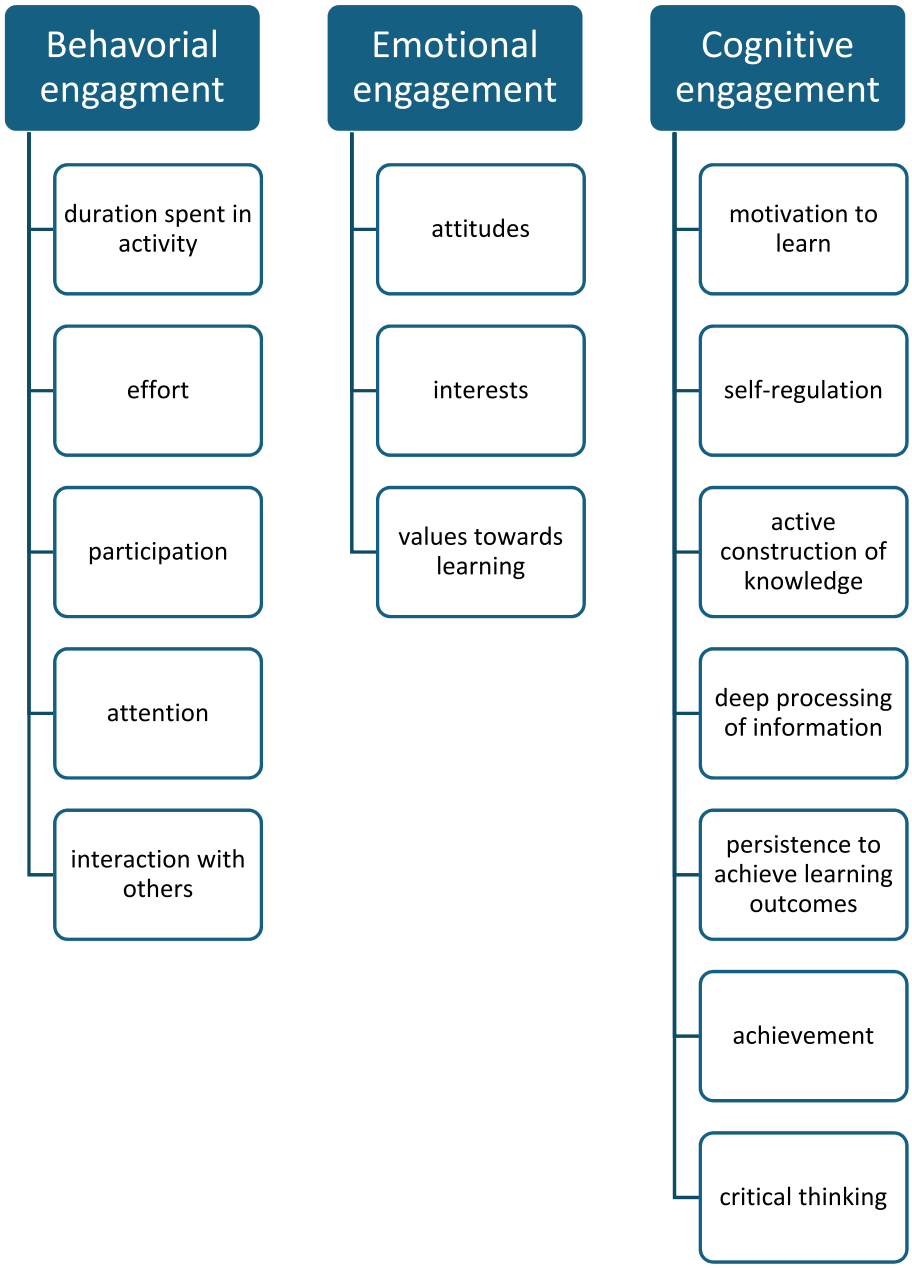



**Fig. 1.** Indicators for behavioral, emotional, and cognitive engagement.

### *2.3. The need for this systematic literature review and meta-analysis*

Some meta-analyses have examined the effects of ChatGPT on student learning. For example, a meta-analysis of 12 early studies in 2022–2023 conducted by Rabiu and Nuhu (2024) showed positive impacts of ChatGPT on the teaching and learning process; other meta-analysis studies also revealed a significant effect of using ChatGPT on learning in language (Wang et al., 2024), medicine (Levin et al., 2024), and education (Li et al., 2024). Li et al. (2024) argued that students require teacher support to engage ChatGPT in learning, supported by an empirical study conducted by Chiu et al. (2024). Moreover, Zhao et al.'s (2023) systematic review suggested how ChatGPT could impact student engagement remain unclear.

To the best of our knowledge, no meta-analyses have examined the impact of ChatGPT on student engagement. This gap is notable given the increasing interest in ChatGPT, which has implications for its application in education to enhance students' learning experience. The absence of comprehensive analyses hinders our understanding of the impact of ChatGPT on students' engagement. By synthesizing existing research, this meta-analysis could provide insights into the roles of ChatGPT in impacting students' engagement in various disciplines across K12 and higher education.

## 3. This study

### *3.1. Research questions*

This systematic literature review and meta-analysis study is to synthesize the available evidence on ChatGPT's impact on student engagement across different educational contexts. We computed the effect size to reveal whether there is a statistical significance between ChatGPT and student engagement. Additionally, this study also used the current literature to suggest directions for future research. The two main research questions are:

RQ1: Does ChatGPT better engage students behaviorally, cognitively, and emotionally than non-ChatGPT?
RQ2: How does ChatGPT foster behavioral, cognitive, and emotional engagement?

We used Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) to identify articles for analysis and present the results (Tricco et al., 2018).

### 3.2. Article selection process

To identify articles for the analysis, we retrieved four widely used digital academic databases related to education: Web of Science, Scopus, Proquest, and ERIC. ChatGPT and engagement are two key components in our research questions, which allowed use to retrieve the relevant articles for this study. Therefore, the search query was "ChatGPT and engagement" in all fields including title, abstract and keywords. During the search process, the inclusive selection criteria are (a) journal articles, (b) written in English, (c) empirical studies, (d) educational research, and (e) published between 2021 and 2024. The exclusive criteria are (a) the intervention is not related to ChatGPT and (b) empirical studies miss key data. The initial and revised search was completed on August 24, 2024. The first three inclusive selection criteria are supported by related meta-analysis studies such as Benavides-Varela et al., 2020 and Zhan et al. (2022).

### 3.3. Article identification

Screening and inclusion criteria were adopted to select articles for the primary analysis. First, 39 duplicate records were removed. Two independent raters then examined the titles and abstracts of the articles to identify articles related to ChatGPT and student engagement. The full texts of the remaining 66 articles were then read to exclude further papers irrelevant to AI or not empirical studies or missing data. Finally, 17 articles were included in the meta-analysis (See Fig. 2 and Table 1). This number of the articles offer reliable analysis which is supported by studies such as 14 articles in Zhan et al. (2022) and 13 articles in Benavides-Varela et al. (2020).

### 3.4. Coding schemes and procedure

This study analyzed the articles in three aspects: behavioral, cognitive, and emotional engagement, i.e., we need to sort the articles into three groups to answer the two research questions. These groups are theoretically driven; therefore, we used a deductive approach to code the articles for analysis (Chiu, 2021, 2022). See the following coding scheme.

- Behavioral engagement was coded by student learning behaviors with ChatGPT. The activities include asking questions, answering prompts, showing their willingness, commitment, and efforts to

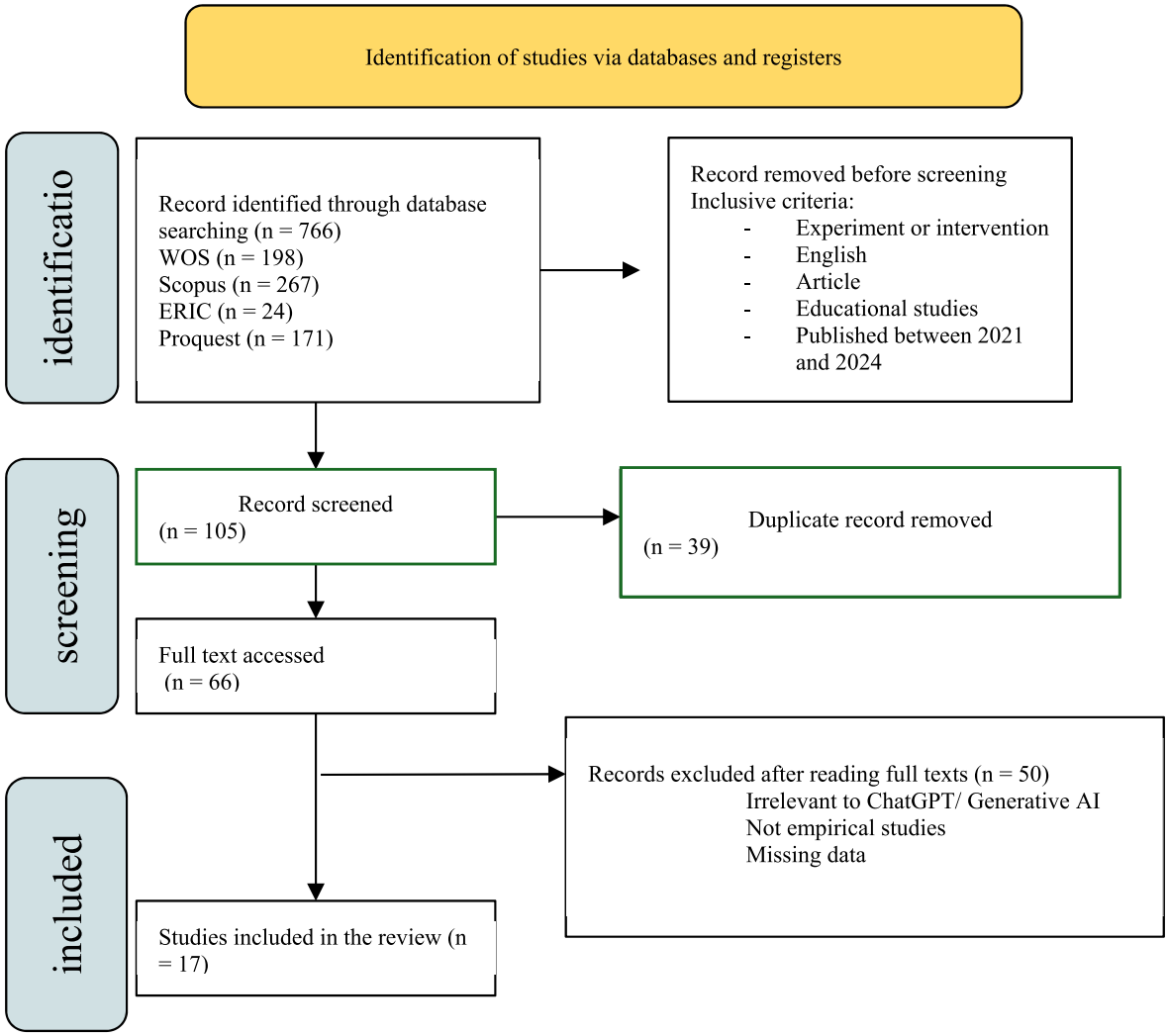



**Fig. 2.** Article selection process following PRISMA approach.

**Table 1**
Identified 17 articles used for meta-analyses.

| Application of ChatGPT in education | N | Studies |
|---|---|---|
| **Higher education** | 13 | |
| **Science and Technology education** | 7 | (Huesca et al., 2024; Jing et al., 2024; Kosar et al., 2024; Qureshi, 2023; Saritepeci & Yildiz Durak, 2024; Sun et al., 2024; Yilmaz et al., 2023) |
| Guidance tool - code optimization and comparison with their code | 2 | (Kosar et al., 2024; Qureshi, 2023) |
| Support problem-solving | 3 | (Jing et al., 2024; Sun et al., 2024; Yilmaz et al., 2023) |
| Support flipped learning | 1 | Huesca et al. (2024) |
| Provide guidance and facilitate collaboration, support creativity | 1 | Saritepeci and Yildiz Durak (2024) |
| **Language learning** | 2 | (Escalante et al., 2023; Pellas, 2023) |
| Generate personalized feedback | 1 | Escalante et al. (2023) |
| Generate digital stories | 1 | Pellas (2023) |
| **Arts education** | 2 | (Bilal Khallel, 2024; Yildiz Durak, 2023) |
| Support problem-solving in case studies | 1 | Bilal Khallel (2024) |
| Guidance tool on visual design tasks | 1 | Yildiz Durak (2023) |
| **Health education** | 2 | (Chang et al., 2023; Dindorf et al., 2024) |
| Guide decision-making to promote students' evaluative skills in an interdisciplinary context | 1 | Dindorf et al. (2024) |
| Support problem-solving; explain disease pathophysiology and provide diagnostic recommendations; | 1 | Chang et al. (2023) |
| **K12 education** | 4 | |
| **Mathematics education** | 1 | Wu et al. (2024) |
| Support self-regulation process | 1 | Wu et al. (2024) |
| **Science education** | 3 | (Alneyadi & Wardat, 2023; Ng et al., 2024; Wardat & Alneyadi, 2024) |
| Personalized instruction | 1 | Ng et al. (2024) |
| Assisted Electronic magnetism learning activity | 1 | Alneyadi and Wardat (2023) |
| Provide recommendations with science-related examples and explanations. Support self-regulation | 1 | Wardat and Alneyadi (2024) |
| **Grand Total** | **17** | |

interact, demonstrating diligence by completing tasks, and adhering to teacher instructions and classroom rules. Other than activities, this engagement can be indicated by the number of task completions and time on task.

- Cognitive engagement was coded by student cognitive learning with ChatGPT. The activities include goal setting, self-reflection, self-improvement, attention to detail, and use of cognitive strategies. Students show their willingness, commitment, and efforts to challenge the output generated,
- Emotional engagement was coded by the emotional psychological measure during learning with ChatGPT. The measures include fun, anxiety, sadness, interest, enjoyment, and frustration.

Two independent coders coded the articles, and a moderator made the final decisions when the two coders had different opinions. The inter-rater coefficient is 0.94.

### 3.5. Computing effect size

We adopted the Comprehensive Meta-Analysis software version 4.0 to compute the effect sizes of the selected studies (Borenstein, 2022). All p-values reported were two-tailed. Heterogeneity was tested with the I2 test, of which heterogeneity can be considered unimportant when it falls between 0% and 40%. In contrast, it is moderate at 30%–60%, significant at 50%–90%, and substantial when it falls between 75% and 100% (Higgins & Thompson, 2002). The random effects were used as we assume that the true effect size can vary across studies due to both random error and genuine differences in effect sizes (Borenstein, 2022). Hedges's g, a statistical method that considers the variability between studies and provides a more accurate estimation of the true effect size, was adopted in the study. Hedges' g values 0.15, 0.40, and 0.75 should be used as thresholds to interpret small, medium, and large effects, respectively (Brydges, 2019). A medium effect size is used to determine if there are significant differences between two groups (Brydges, 2019). It allows for more accurate comparisons between studies with different sample sizes (Bai et al., 2020; Korpershoek et al., 2016). We used various statistical results from the studies, including means, standard deviations, and standardized mean differences (estimated with T-tests or F-tests), as the input for effect size calculations.

We computed the effect size for each eligible study to fit the assumption of independence that the data of different outcomes were derived from different samples (Hedges et al., 2010). It could be problematic to consider the outcomes of the same set of samples as independent, as this could lead to an erroneously small standard error and a narrow confidence interval for the point estimates computed across outcomes. Nevertheless, if a study involved different sets of samples without overlap, we computed one effect size for each sample set. If a study had multiple dimensions of student engagement reflected from the same set of samples, we treated their sub-studies as distinct studies to allow for a more nuanced understanding of the effects being measured (Borenstein & Higgins, 2013). For instance, while reporting impacts on students' learning, the measurement was chosen across three dimensions of student engagement: cognitive, behavioral, and emotional (Fredricks et al., 2004).

### 3.6. Publication bias

Publication bias arises from the tendency of researchers, reviewers, and editors to favor the publication of positive research findings (Turner, 2013). Publication bias is reflected when the potential distortion of treatment effects was assessed with Funnel plots using Hedges's g with random effects (Duval & Tweedie, 2000). In an unbiased scenario, the plot should display a symmetrical inverted funnel. Conversely, if publication bias is present, the plot will exhibit asymmetry and skewness. Suppose the plot appears somewhat asymmetric, with more studies on the right side of the mean effect size (positive side). In that case, it implies a potential publication bias, as negative or non-significant studies might be underrepresented.

To address the research questions regarding student engagement, the included studies' findings and discussions were analyzed thematically using an inductive thematic approach by two independent raters. The general framework for the thematic analysis followed the three-dimensional model of student engagement (Fredricks et al., 2004), involving behavioral, emotional, and cognitive engagement.

## 4. Results and discussions

### 4.1. Descriptive statistics

Table 1 shows the 17 selected articles for analysis. The majority of the articles focused on higher education (n = 13; 76%), while a quarter of studies focused on K-12 education (n = 4; 24%). The articles covered diverse subject disciplines, including science and technology education (n = 10; 58.8 %), health education (n = 2; 11.8%), arts education (n = 2; 11.8%), language learning (n = 2; 11.8%), and mathematics education (n = 1; 5.9%). As shown in Fig. 3, the 17 selected studies took place on three continents: Asia (n = 10; 59%), Europe (n = 6; 35%), and North America (n = 1; 6%). Overall, most of the selected articles examined the effects of ChatGPT on student engagement in STEAM contexts in higher education. They also capture views from both Eastern and Western cultures.

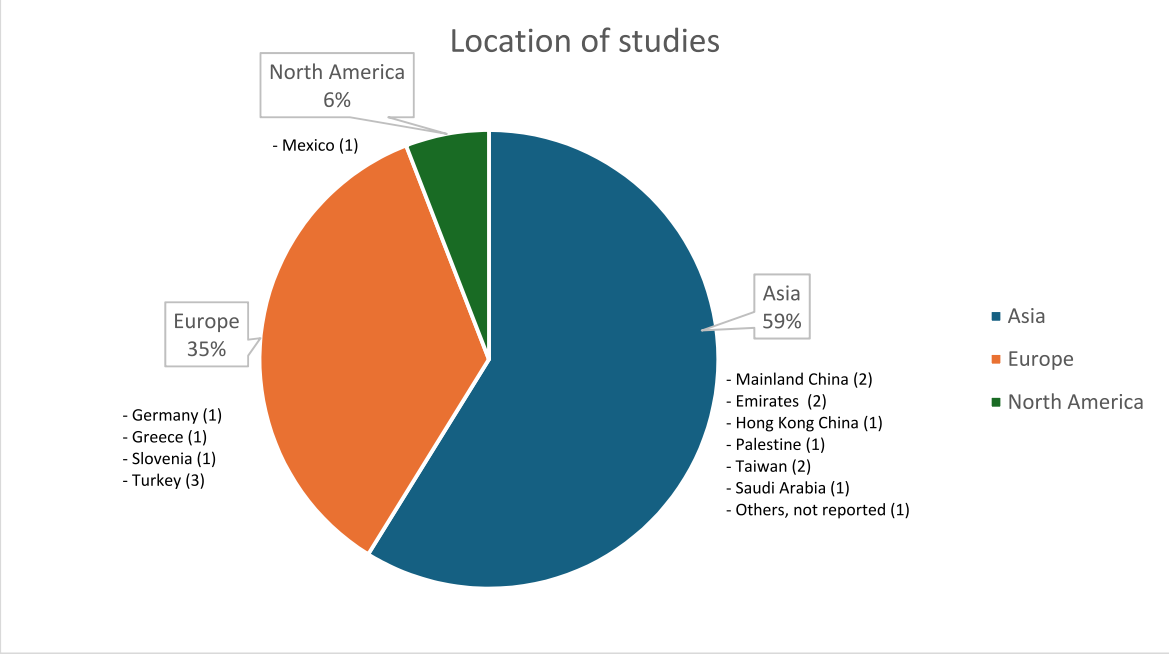



**Fig. 3.** Location of studies.

### *4.2. Does ChatGPT better engage students behaviorally, cognitively, and emotionally than non-ChatGPT?*

#### *4.2.1. Methodological quality*

We evaluated the methodological quality of experimental and quasi-experimental research studies using the Medical Education Research Study Quality Instrument (MERSQI) (Reed et al., 2008). The ten items reflect six domains of study quality: study design, sampling, data type, validity of assessments, data analysis, and outcomes. The possible MERSQI scores range from 5 to 18. The MERSQI items presented in the eligible studies are listed in Table 2. The MERSQI is designed for quantitative studies (n = 17) and its mean score (M = 10.15) of the quantitative studies) falls within the published range of other quantitative studies for technology-incorporated learning activities (Bai et al., 2020; Weng et al., 2023).

**Table 2**
Methodology quality result using MERSQI.

| Domain | MERSQI items presented (score) | N | Percentages |
|---|---|---|---|
| Study design | Non-randomized (2) | 11 | 64.71 |
| | Randomized control trial (3) | 6 | 35.29 |
| Sampling | Institution studied | – | – |
| | One institution studied (0.5) | 15 | 88.24 |
| | Two institutions studied (1) | 2 | 11.76 |
| | Response rate | – | – |
| | <50% or not specified (0.5) | 3 | 17.65 |
| | 50–74% (1) | 6 | 35.29 |
| | ≥75 (1.5) | 8 | 47.06 |
| Type of data | Assessed by participants (1) | 10 | 58.82 |
| | Objective measurement (3) | 7 | 41.18 |
| Validity evidence for evaluation instrument scores | Content of instrument Reported (1) | – 17 | – 100 |
| Data analysis | Appropriateness of analysis | – | – |
| | Appropriate for study design, type of data (1) | 17 | 100 |
| | Complexity of analysis | – | – |
| | Beyond descriptive analysis (2) | 17 | 100 |
| Outcomes | attitudes, perceptions (1) | 5 | 29.41 |
| | Knowledge, skills (1.5) | 12 | 70.59 |

We analyzed 17 articles to determine the effects of ChatGPT and non-ChatGPT on student engagement. Effect sizes for each study were calculated. The Q statistics value suggested the presence of heterogeneity ($Q = 126.981$, $I2 = 79\%$, $p < 0.001$). With the random effects model, see Fig. 4, a medium effect size of Hedges's $g = 0.55$ was obtained ($p < 0.05$) (Sullivan & Feinn, 2012). This indicated that students in ChatGPT-mediated learning activities had gained significantly more engagement levels than those involved in non-ChatGPT learning activities.

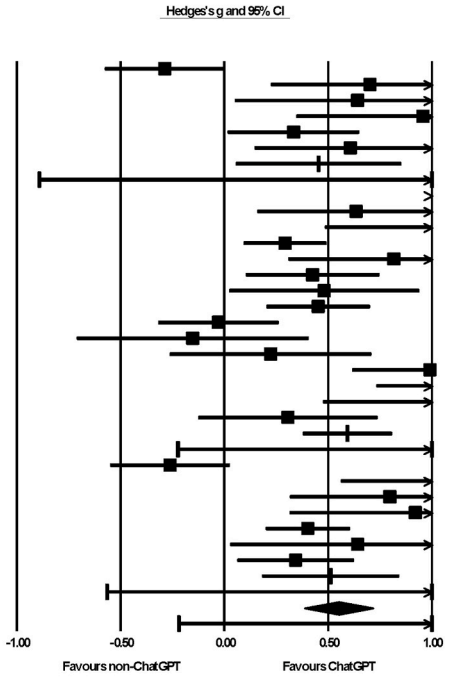
#### *4.2.2. Behavioral engagement*

Six studies involving 435 participants were analyzed to investigate the effects of ChatGPT and non-ChatGPT interventions on student behavioral engagement. Six independent effect sizes were calculated. The Q statistic value indicated the presence of significant heterogeneity among the studies ($Q = 26.142$, $I^2 = 81\%$, $p < 0.001$). Using a random effects model (Fig. 5), an overall medium effect size of Hedges's g was 0.454 (95% CI = 0.055 to 0.853) (Sullivan & Feinn, 2012), indicating that students participating in ChatGPT-mediated learning activities exhibited significantly greater behavioral engagement compared to those engaged in non-ChatGPT learning activities.

Despite the overall positive findings, the funnel plot analysis (Fig. 6) appeared relatively asymmetric, suggesting potential publication bias (Duval & Tweedie, 2000). Kendall's Tau calculation (tau = 0.194, p-value = 0.44) indicated a weak positive correlation with a non-significant p-value, implying no strong statistical evidence for publication bias, although visual inspection suggests caution.

Among the six studies that examined the impact of ChatGPT on student behavioral engagement, five reported a statistically significant positive effect (Dindorf et al., 2024; Ng et al., 2024; Wu et al., 2024; Yilmaz et al., 2023), while one study indicated that ChatGPT did not significantly improve students' behavioral engagement compared to a control group at certain points in time (Saritepeci & Yildiz Durak, 2024). However, their findings still showed evidence of overall improvement in reflective thinking for both the experimental and control groups over time. Furthermore, it has been demonstrated that ChatGPT has enhanced students' help-seeking behaviors in science education (Ng et al., 2024), promoted more proactive behaviors in learning, and improving student behavioral engagement in self-regulation

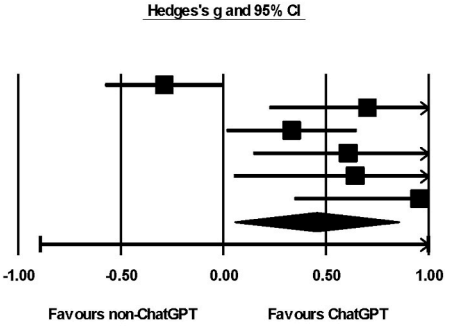
| Group by Subgroup within study | Studyname | Subgroup within study | Statistics for each study | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Hedges's g | Standard error | Variance | Lower limit | Upper limit | Z-Value | p-Value |
| behavioural | Saritepeci & Yildiz Durak, 2024 | behavioural | -0.288 | 0.147 | 0.022 | -0.576 | 0.001 | -1.954 | 0.051 |
| behavioural | Wu et al., 2024 | behavioural | 0.703 | 0.244 | 0.059 | 0.225 | 1.181 | 2.884 | 0.004 |
| behavioural | Yilmaz & Karaoglan Yilmaz, 2023 | behavioural | 0.643 | 0.301 | 0.091 | 0.052 | 1.233 | 2.134 | 0.033 |
| behavioural | Yilmaz, 2023 | behavioural | 0.957 | 0.311 | 0.097 | 0.347 | 1.567 | 3.077 | 0.002 |
| behavioural | Dindorf et al., 2024 | behavioural | 0.334 | 0.161 | 0.026 | 0.017 | 0.650 | 2.068 | 0.039 |
| behavioural | Ng et al., 2024 | behavioural | 0.609 | 0.235 | 0.055 | 0.147 | 1.070 | 2.585 | 0.010 |
| behavioural | Pooled | | 0.454 | 0.204 | 0.041 | 0.055 | 0.853 | 2.232 | 0.026 |
| behavioural | Prediction Interval | | 0.454 | | | -0.892 | 1.800 | | |
| cognitive | Chang et al., 2023 | cognitive | 1.695 | 0.326 | 0.106 | 1.056 | 2.333 | 5.199 | 0.000 |
| cognitive | Wu et al., 2024 | cognitive | 0.635 | 0.242 | 0.059 | 0.160 | 1.110 | 2.619 | 0.009 |
| cognitive | Yilmaz & Karaoglan Yilmaz, 2023 | cognitive | 1.104 | 0.316 | 0.100 | 0.485 | 1.723 | 3.496 | 0.000 |
| cognitive | Bilal Khallel, 2024 | cognitive | 0.292 | 0.101 | 0.010 | 0.094 | 0.491 | 2.885 | 0.004 |
| cognitive | Yilmaz, 2023 | cognitive | 0.818 | 0.259 | 0.067 | 0.309 | 1.326 | 3.151 | 0.002 |
| cognitive | Dindorf et al., 2024 | cognitive | 0.426 | 0.164 | 0.027 | 0.104 | 0.747 | 2.592 | 0.010 |
| cognitive | Ng et al., 2024 | cognitive | 0.481 | 0.233 | 0.054 | 0.024 | 0.939 | 2.061 | 0.039 |
| cognitive | Huesca et al., 2024 | cognitive | 0.453 | 0.127 | 0.016 | 0.203 | 0.702 | 3.555 | 0.000 |
| cognitive | Kosar et al., 2024 | cognitive | -0.029 | 0.148 | 0.022 | -0.318 | 0.260 | -0.196 | 0.844 |
| cognitive | Escalante et al., 2023 | cognitive | -0.153 | 0.285 | 0.081 | -0.711 | 0.405 | -0.538 | 0.591 |
| cognitive | Pellas, 2023 | cognitive | 0.223 | 0.248 | 0.061 | -0.262 | 0.709 | 0.900 | 0.368 |
| cognitive | Alneyadi & Wardat, 2023 | cognitive | 0.991 | 0.191 | 0.036 | 0.617 | 1.365 | 5.189 | 0.000 |
| cognitive | Wardat & Alneyadi, 2024 | cognitive | 1.130 | 0.202 | 0.041 | 0.733 | 1.527 | 5.585 | 0.000 |
| cognitive | Qureshi, 2023 | cognitive | 1.073 | 0.304 | 0.093 | 0.476 | 1.669 | 3.526 | 0.000 |
| cognitive | Sun et al.,2024 | cognitive | 0.306 | 0.220 | 0.049 | -0.125 | 0.738 | 1.391 | 0.164 |
| cognitive | Pooled | | 0.593 | 0.110 | 0.012 | 0.378 | 0.809 | 5.394 | 0.000 |
| cognitive | Prediction Interval | | 0.593 | | | -0.224 | 1.410 | | |
| emotional | Saritepeci & Yildiz Durak, 2024 | emotional | -0.262 | 0.147 | 0.022 | -0.550 | 0.026 | -1.782 | 0.075 |
| emotional | Chang et al., 2023 | emotional | 1.153 | 0.301 | 0.091 | 0.562 | 1.744 | 3.826 | 0.000 |
| emotional | Wu et al., 2024 | emotional | 0.798 | 0.246 | 0.060 | 0.316 | 1.280 | 3.245 | 0.001 |
| emotional | Yilmaz & Karaoglan Yilmaz, 2023 | emotional | 0.921 | 0.309 | 0.096 | 0.315 | 1.527 | 2.979 | 0.003 |
| emotional | Bilal Khallel, 2024 | emotional | 0.402 | 0.103 | 0.011 | 0.200 | 0.605 | 3.898 | 0.000 |
| emotional | Yilmaz, 2023 | emotional | 0.643 | 0.314 | 0.098 | 0.029 | 1.258 | 2.051 | 0.040 |
| emotional | Jing et al.,2024 | emotional | 0.344 | 0.143 | 0.021 | 0.063 | 0.625 | 2.399 | 0.016 |
| emotional | Pooled | | 0.512 | 0.168 | 0.028 | 0.183 | 0.842 | 3.047 | 0.002 |
| emotional | Prediction Interval | | 0.512 | | | -0.566 | 1.590 | | |
| Overall | Pooled | | 0.550 | 0.084 | 0.007 | 0.385 | 0.714 | 6.553 | 0.000 |
| Overall | Prediction Interval | | 0.550 | | | -0.219 | 1.318 | | |




Meta Analysis

**Fig. 4.** Forest plot of overall student engagement.

| Study name | Subgroup within study | Statistics for each study | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | Hedges's g | Standard error | Variance | Lower limit | Upper limit | Z-Value | p-Value |
| Saritepeci & Yildiz Durak, 2024 | behavioural | -0.288 | 0.147 | 0.022 | -0.576 | 0.001 | -1.954 | 0.051 |
| Wu et al., 2024 | behavioural | 0.703 | 0.244 | 0.059 | 0.225 | 1.181 | 2.884 | 0.004 |
| Dindorf et al., 2024 | behavioural | 0.334 | 0.161 | 0.026 | 0.017 | 0.650 | 2.068 | 0.039 |
| Ng et al., 2024 | behavioural | 0.609 | 0.235 | 0.055 | 0.147 | 1.070 | 2.585 | 0.010 |
| Yilmaz & Karaoglan Yilmaz, 2023 | behavioural | 0.643 | 0.301 | 0.091 | 0.052 | 1.233 | 2.134 | 0.033 |
| Yilmaz, 2023 | behavioural | 0.957 | 0.311 | 0.097 | 0.347 | 1.567 | 3.077 | 0.002 |
| Pooled | | 0.454 | 0.204 | 0.041 | 0.055 | 0.853 | 2.232 | 0.026 |
| Prediction Interval | | 0.454 | | | -0.892 | 1.800 | | |




Meta Analysis

**Fig. 5.** Forest plot of behavioral engagement.

performance stage (Wu et al., 2024); facilitated querying behaviors to develop students' decision-making competencies (Dindorf et al., 2024); facilitated interaction with other people and fostered active participation (Yildiz Durak, 2023; Yilmaz et al., 2023).

### 4.2.3. Cognitive engagement

Fifteen of the collected studies with 1684 participants reported impacts on cognitive engagement. The Q statistics value suggested the presence of heterogeneity ($Q = 63.272$, $I2 = 78\%$, $p < 0.001$). With the random effect model, see Fig. 7, an overall moderate to large effect size of Hedges's $g = 0.593$ (95% CI = 0.378 to 0.809) (Sullivan & Feinn, 2012). Evidently, using ChatGPT in learning activities significantly impacted student cognitive engagement compared to their counterparts in non-ChatGPT learning activities.

The funnel plot appears relatively symmetric, indicating a potential absence of significant publication bias, see Fig. 8. However, the Begg and Mazumdar rank correlation analysis yielded a Spearman's rank correlation coefficient (ρ) of 0.625, suggesting a moderate positive correlation that could imply some level of publication bias in the meta-analysis of cognitive engagement. Despite this moderate correlation, the visual inspection of the funnel plot indicates reasonable symmetry, suggesting no strong evidence of publication bias.

Among the fifteen studies that examined impacts on student cognitive engagement, four of the studies reported no significant cognitive engagement impacts in terms of programming performance (Kosar et al., 2024; Sun et al., 2024), writing proficiency and efficacy (Yildiz Durak, 2023; Yilmaz et al., 2023). The results indicate that while learners in the ChatGPT mode had a higher average score than learners in the non-ChatGPT activities in tests, the scores were not statistically significant. Therefore, the findings highlight the inconsistent cognitive engagement effects between ChatGPT activities and those not involving ChatGPT.

Moreover, there was research evidence that ChatGPT activities can enhance students' (i) academic achievements (Bilal Khallel, 2024; Jing

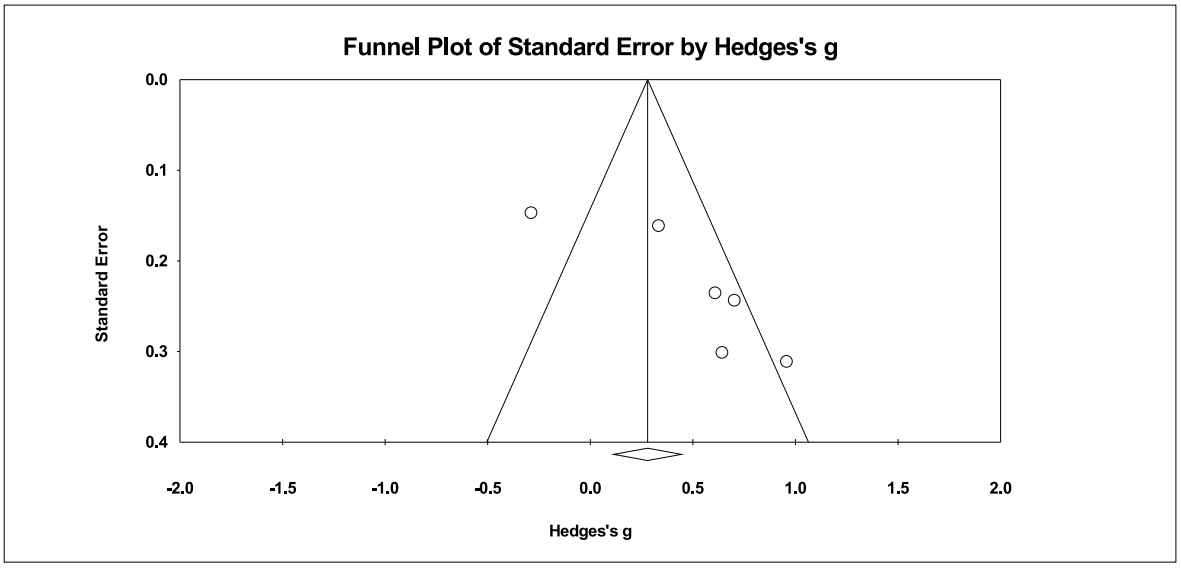



**Fig. 6.** Funnel plot of behavioral engagement.

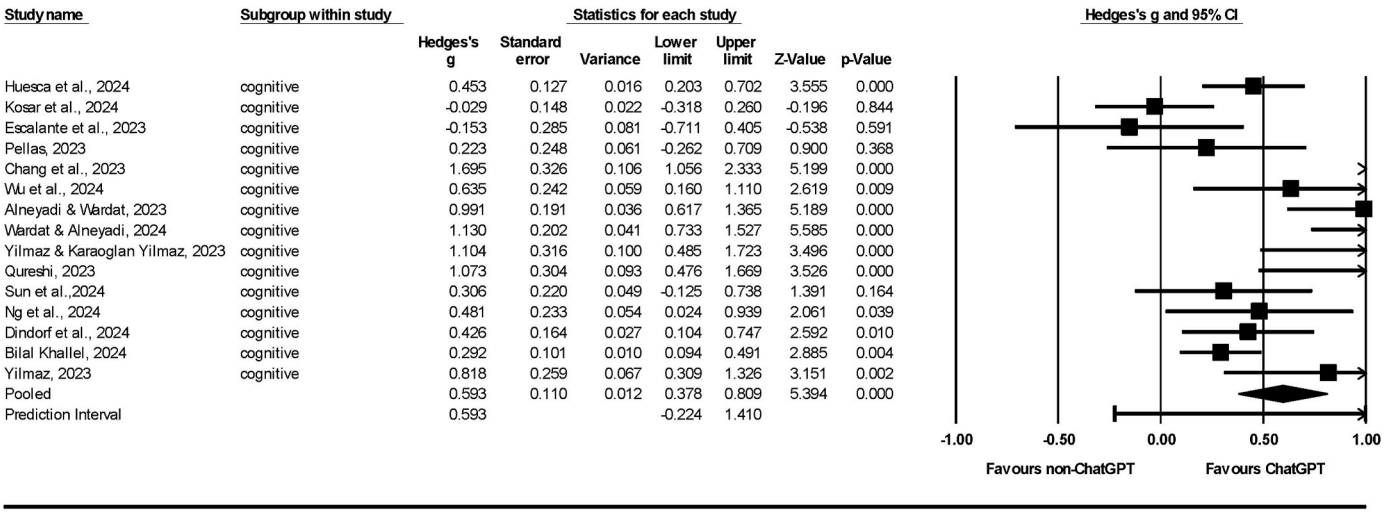



**Fig. 7.** Forest plot of cognitive engagement.

et al., 2024), (ii) self-efficacy in critical thinking, decision-making and visual design (Bilal Khallel, 2024; Jing et al., 2024), (iii) self-regulations and motivation (Huesca et al., 2024; Kosar et al., 2024; Sun et al., 2024).

In science and technology education, ChatGPT can enhance flipped learning by building student programming knowledge and offering guidance on code optimization, code comparison, problem-solving, and promoting collaboration in project-based learning (Huesca et al., 2024; Jing et al., 2024; Kosar et al., 2024; Qureshi, 2023; Sun et al., 2024; Yilmaz et al., 2023). ChatGPT not only provides support and fosters collaboration but also inspires creativity among students by suggesting topics for collaborative story writing, guiding them in scriptwriting, and integrating other generative AI tools like Mid-journey for creating scripted scenes and voice acting (Saritepeci & Yildiz Durak, 2024). In language education, ChatGPT offers personalized feedback to assist English academic writing and collaborates with students in crafting creative digital stories (Escalante et al., 2023; Pellas, 2023). In arts education, ChatGPT aids problem-solving within the context of disease pathophysiology and provides diagnostic recommendations, which enhance decision-making and foster critical thinking skills across multiple disciplines (Chang et al., 2023; Dindorf et al., 2024). For K12 education, ChatGPT promotes self-regulation processes in mathematics education through flipped-blended learning (Wu et al., 2024). It records students' questions and answers while engaging them actively to alleviate anxiety over unresolved questions, thus reducing interruptions during the self-regulation performance phase and encouraging self-reflection. Similarly, in science education, ChatGPT's personalized instructions and recommendations, with some science-related examples and explanations, support students' self-regulation in their science learning activities (Alneyadi & Wardat, 2023; Ng et al., 2024; Wardat & Alneyadi, 2024).

#### 4.2.4. *Emotional engagement*

Seven of the collected studies (e.g., with 538 participants) reported impacts on emotional engagement. The Q statistics value suggested the presence of heterogeneity ($Q = 32.289$, $I2 = 81\%$, $p < 0.001$). With the random effect model, see Fig. 9, a medium effect size of Hedges's $g = 0.512$ (95%CI = 0.183 to 0.842) (Sullivan & Feinn, 2012). Evidently, the adoption of ChatGPT in learning activities significantly impacted

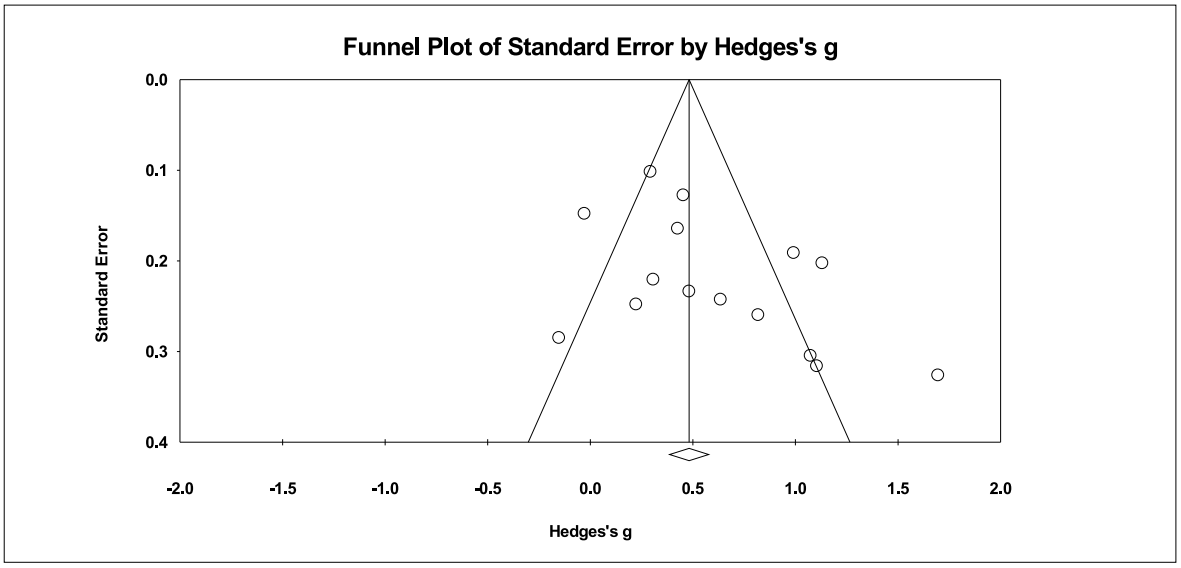
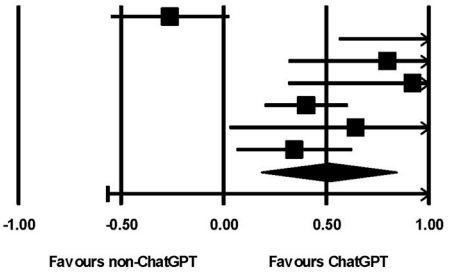



**Fig. 8.** Funnel plot of cognitive engagement.

| Study name | Subgroup within study | Hedges's g | Standard error | Variance | Lower limit | Upper limit | Z-Value | p-Value |
|---|---|---|---|---|---|---|---|---|
| Saritepeci & Yildiz Durak, 2024 | emotional | -0.262 | 0.147 | 0.022 | -0.550 | 0.026 | -1.782 | 0.075 |
| Chang et al., 2023 | emotional | 1.153 | 0.301 | 0.091 | 0.562 | 1.744 | 3.826 | 0.000 |
| Wu et al., 2024 | emotional | 0.798 | 0.246 | 0.060 | 0.316 | 1.280 | 3.245 | 0.001 |
| Yilmaz & Karaoglan Yilmaz, 2023 | emotional | 0.921 | 0.309 | 0.096 | 0.315 | 1.527 | 2.979 | 0.003 |
| Bilal Khallel, 2024 | emotional | 0.402 | 0.103 | 0.011 | 0.200 | 0.605 | 3.898 | 0.000 |
| Yilmaz, 2023 | emotional | 0.643 | 0.314 | 0.098 | 0.029 | 1.258 | 2.051 | 0.040 |
| Jing et al.,2024 | emotional | 0.344 | 0.143 | 0.021 | 0.063 | 0.625 | 2.399 | 0.016 |
| Pooled | | 0.512 | 0.168 | 0.028 | 0.183 | 0.842 | 3.047 | 0.002 |
| Prediction Interval | | 0.512 | | | -0.566 | 1.590 | | |




Meta Analysis

**Fig. 9.** Forest plot of emotional engagement.

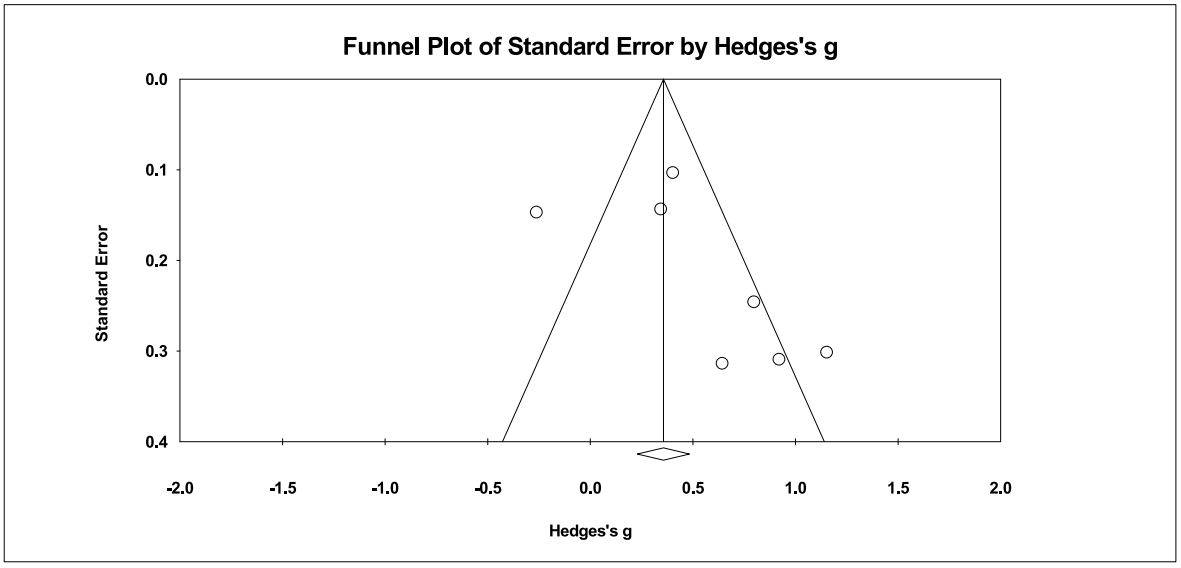



**Fig. 10.** Funnel plot of emotional engagement.

students' emotional engagement compared to their counterparts in non-ChatGPT learning activities.

The funnel plot appears relatively asymmetric, indicating the presence of publication bias, as shown in Fig. 10. (Duval & Tweedie, 2000). The Begg and Mazumdar rank correlation was calculated to check the symmetry, yielding a Kendall's Tau value of 0.4286 with a p-value of 0.20, suggesting a moderate positive correlation but not statistically significant. This indicates no strong evidence of publication bias in the meta-analysis of ChatGPT's impact on students' emotional engagement. However, the visual inspection of the funnel plot still suggests some asymmetry, implying that caution is warranted in interpreting the results.

Among the seven studies that examined impacts on student emotional engagement, one of the studies reported no significant emotional engagement impacts in terms of comfortability with problems (Saritepeci & Yildiz Durak, 2024); this demonstrated the inconsistent impact on student engagement between ChatGPT and non-ChatGPT activities. More prominently, there was research evidence that ChatGPT activities can enhance students' positive attitudes and expectations towards learning (Yildiz Durak, 2023; Yilmaz et al., 2023), in terms of (1) enjoyment (Bilal Khallel, 2024; Jing et al., 2024); (2) intention to utilize technology (Bilal Khallel, 2024; Jing et al., 2024). Qualitative findings also revealed an increase in students' interest and confidence in learning with ChatGPT (Huesca et al., 2024; Kosar et al., 2024; Sun et al., 2024). Furthermore, Bilal Khallel (2024) added that participants in the peer learning group had a better perception of online classes' satisfaction and usefulness than the participants in the independent and group learning environments. Therefore, integrating ChatGPT into online peer group settings can be beneficial in enhancing emotional engagement through customized interactions, adaptable content delivery, and enhanced engagement levels.

### 4.3. How does ChatGPT foster behavioral, cognitive, and emotional engagement?

A summary of student engagement reported in the collected studies can be synthesized into three dimensions, see Table 3: (i) behavioral engagement (n = 20, 30%), which encompasses students' participation, attention, and effort; (ii) cognitive engagement (n = 30, 45%), which reflects the level of investment in learning and mastering content; (iii) emotional engagement (n = 17, 25%), which relates to the emotions and reactions that facilitate task completion (Christenson et al., 2012; Fredricks et al., 2004; Kahu, 2013).

We identified three main roles of ChatGPT, see Fig. 11, in supporting student learning, namely personalized tutoring (PT) (n = 10, 59 %), programming and technical assistance (TA) (n = 5, 29%), and content generation and collaboration (CG) (n = 2, 12%). Specifically, ChatGPT supports personalized tutoring by facilitating engaging dialogues, offering informative responses, and delivering personalized supplementary materials that cater to each student's diverse learning needs (Dindorf et al., 2024). For example, it has been shown to increase student engagement by providing instant feedback and clarification on a diverse range of tasks, including decision-making (n = 1), English writing (n = 1), problem-solving tasks (n = 2), constructing programming conceptual knowledge (n = 1), visual design tasks (n = 1), providing science learning instruction and recommendations (n = 3), and mathematics learning tasks (n = 1).

#### 4.3.1. Behavioral engagement

As shown in Table 3 and Fig. 11, PT, TA, and CG impact behavioral engagement. ChatGPT encourages students to participate more actively in their learning processes, as they can seek help without hesitation (PT, TA, and CG). Therefore, students learn more proactively as the support from ChatGPT scaffolds their learning, preventing them from interrupting their learning due to unresolved problems (Sun et al., 2024; Wu et al., 2024; Yildiz Durak, 2023). Specifically, in programming education, students leverage ChatGPT to optimize and compare code (TA) (Kosar et al., 2024; Sun et al., 2024). If prompted correctly, ChatGPT offers step-by-step guidance explaining the underlying algorithm to be used in the solution with code snippets to support problem-solving (Qureshi, 2023). However, it was found that students frequently copy and paste codes from ChatGPT when reading, debugging, and solving programming tasks (Sun et al., 2024). It shows that students' overreliance on ChatGPT may disrupt their participation in learning. Even with the help of ChatGPT, students' intention to complete challenging programming questions did not increase (Yilmaz et al., 2023). Apart from students' interaction with the chatbot, there is a two-sided finding regarding students' interaction with others. Some students actively engage in pre-class activities with their classmates when ChatGPT guides their collaboration (CG) (Huesca et al., 2024; Saritepeci & Yildiz Durak, 2024), while some students raised their concerns about ChatGPT's impact on reduced human interaction among their peers and human tutors (Dindorf et al., 2024; Escalante et al., 2023). Overall, the impact on behavioral engagement was evident in the enhancement of students' participation (n = 30) and the facilitation of interaction with peers (n = 2), as noted in studies by Huesca et al. (2024) and Saritepeci and Yildiz Durak (2024). However, a reduction in interaction with others was also observed in two studies (n = 2) (Dindorf

**Table 3**
Distributions of behavioral, cognitive and emotional engagement in the selected studies.

| Student Engagement | N | |
|---|---|---|
| **Behavioral engagement** | **20** | |
| Interaction with others | 2 | (Dindorf et al., 2024; Huesca et al., 2024) |
| Participation | 15 | (Alneyadi & Wardat, 2023; Bilal Khallel, 2024; Chang et al., 2023; Dindorf et al., 2024; Huesca et al., 2024; Jing et al., 2024; Kosar et al., 2024; Ng et al., 2024; Pellas, 2023; Saritepeci & Yildiz Durak, 2024; Sun et al., 2024; Wardat & Alneyadi, 2024; Wu et al., 2024; Yildiz Durak, 2023; Yilmaz et al., 2023) |
| Reduce interaction with others | 2 | (Dindorf et al., 2024; Escalante et al., 2023) |
| Overreliance | 1 | Dindorf et al. (2024) |
| **Cognitive engagement** | **30** | |
| Achievement | 3 | (Alneyadi & Wardat, 2023; Huesca et al., 2024; Qureshi, 2023) |
| Active construction of knowledge | 4 | (Dindorf et al., 2024; Huesca et al., 2024; Jing et al., 2024; Wu et al., 2024) |
| Critical thinking | 9 | (Chang et al., 2023; Dindorf et al., 2024; Huesca et al., 2024; Jing et al., 2024; Saritepeci & Yildiz Durak, 2024; Sun et al., 2024; Wardat & Alneyadi, 2024; Yildiz Durak, 2023) |
| Deep processing of information | 3 | (Alneyadi & Wardat, 2023; Dindorf et al., 2024; Yilmaz et al., 2023) |
| Motivation | 3 | (Kosar et al., 2024; Ng et al., 2024; Yilmaz et al., 2023) |
| Reflective thinking | 1 | Saritepeci and Yildiz Durak (2024) |
| Self-regulation | 6 | (Bilal Khallel, 2024; Ng et al., 2024; Pellas, 2023; Wardat & Alneyadi, 2024; Wu et al., 2024; Yildiz Durak, 2023) |
| Weaken critical thinking | 1 | Sun et al. (2024) |
| **Emotional engagement** | **17** | |
| Confidence | 1 | Bilal Khallel (2024) |
| Enjoyment | 2 | (Chang et al., 2023; Wu et al., 2024) |
| Frustration | 1 | Alneyadi and Wardat (2023) |
| Hesitation | 1 | Alneyadi and Wardat (2023) |
| Intention to use | 1 | Jing et al. (2024) |
| Interest | 6 | (Alneyadi & Wardat, 2023; Huesca et al., 2024; Jing et al., 2024; Kosar et al., 2024; Saritepeci & Yildiz Durak, 2024; Sun et al., 2024) |
| Satisfaction | 4 | (Alneyadi & Wardat, 2023; Bilal Khallel, 2024; Escalante et al., 2023; Yildiz Durak, 2023) |
| Reduced frustration | 1 | Wu et al. (2024) |

*Remark: One or more student engagement impacts could be identified in one paper.

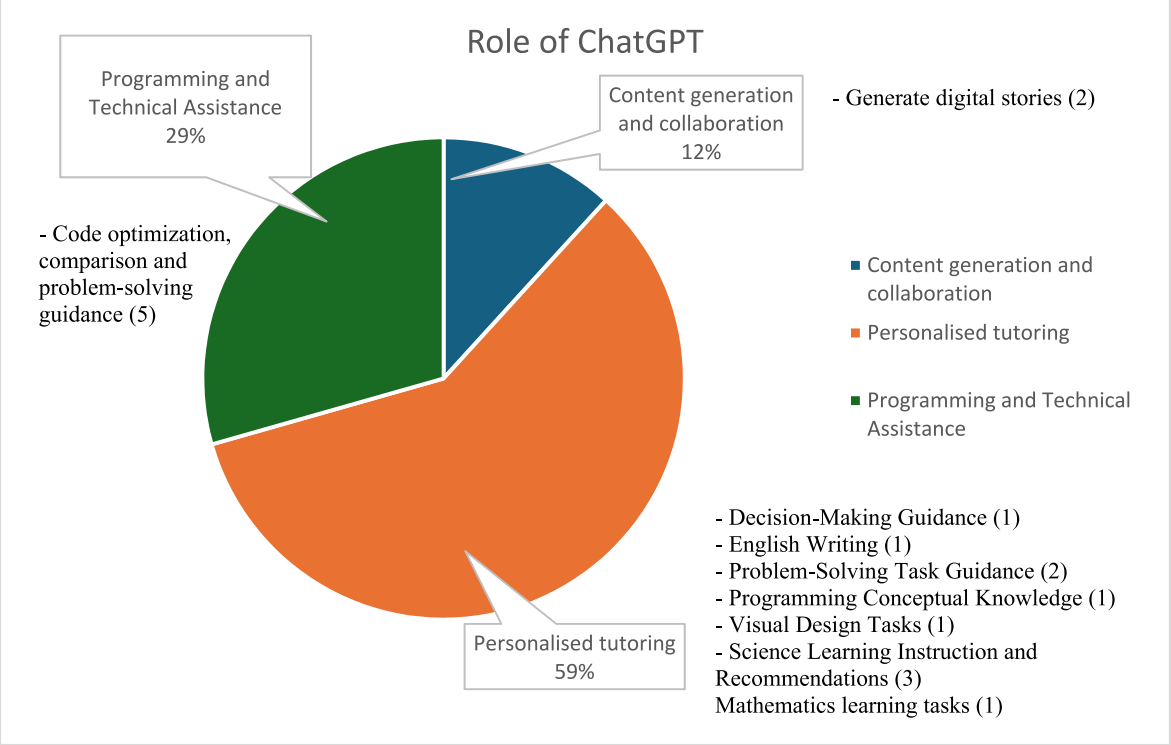



**Fig. 11.** Role of ChatGPT.

et al., 2024; Escalante et al., 2023). Besides, Dindorf et al., 2024 also raised concerns about the overreliance on the tool.

#### 4.3.2. Cognitive engagement

As showed in Table 3 and Fig. 11, PT, TA and CG impact cognitive engagement in terms of achievement (n = 3), active construction of knowledge (n = 4), confidence (n = 1), critical thinking (n = 9), deep processing of information (n = 3), motivation (n = 4), reflective thinking (n = 1), satisfaction (n = 4), self-regulation (n = 6), and instances of weakened critical thinking (n = 1) (Sun et al., 2024). Studies highlight various aspects of how students cognitively interact with ChatGPT (CG), with critical thinking being the most frequently addressed learning outcomes (Alneyadi & Wardat, 2023; Chang et al., 2023; Saritepeci & Yildiz Durak, 2024; Yildiz Durak, 2023). Although some areas, such as confidence and reflective thinking, were less represented, they still play a crucial role in overall cognitive engagement (Saritepeci & Yildiz Durak, 2024). For example, ChatGPT alleviated students' struggles in the performance stage of self-regulated learning by providing convergent information (PT), supporting their knowledge construction (TA), and boosting their cognitive engagement (Bilal Khallel, 2024; Ng et al., 2024; Wu et al., 2024). A respondent suggested that ChatGPT was a virtual teacher by providing personalized suggestions and reminders to keep them motivated and organized in achieving their learning goals (PT) (Ng et al., 2024). The chatbot guided them on effective note-taking and task prioritization, providing training questions on disciplinary topics with numerous examples and explanations for knowledge checking (PT). The guidance provided by ChatGPT was useful for solving complex problems and learning unfamiliar topics (Kosar et al., 2024; Sun et al., 2024; Wardat & Alneyadi, 2024). For example, ChatGPT broadened students' understanding of the subject beyond the theoretical concepts, inspiring them to reflect on the importance of Quantum Theory in real-life applications (PT and TA). ChatGPT also developed students' critical thinking and problem-solving abilities by giving clear explanations and examples of domain-specific problems in situational contexts, supporting students in acquiring essential knowledge for constructing cognitive concepts through problem-solving, followed by promoting further searches and inquiries for self-regulated learning (Chang et al., 2023; Huesca et al., 2024; Sun et al., 2024).

#### 4.3.3. Emotional engagement

PT provided by ChatGPT had an impact on student emotional engagement. As shown in Table 3, the selected studies captured both positive and negative effects of ChatGPT on student emotional engagement, revealing impacts on their confidence (n = 1), enjoyment (n = 2), interests (n = 6), intention to use (n = 1), and positive attitudes and expectations (n = 3), as highlighted by Jing et al. (2024) and Kosar et al. (2024). Additionally, one study identified instances of student hesitation (n = 1), as reported by Alneyadi and Wardat (2023). Compared to traditional teaching methods, which emphasize delivering teaching content and offering limited interactions between teachers and students and delayed feedback, ChatGPT provides adaptable content with real-time feedback catering for different learning levels of students (PT) so they reduced frustration and showed higher learning enjoyment, positive attitude on learning, and greater intention to use technology, and higher confidence when they learned efficiently and timely with ChatGPT through open-ended chats (Chang et al., 2023; Jing et al., 2024; Kosar et al., 2024; Yildiz Durak, 2023; Yilmaz et al., 2023).

## 5. Conclusion, limitations, and future directions

### 5.1. Overall implications

We analyzed 17 articles to synthesize their data to understand better how ChatGPT-incorporated learning activities impact student behavioral, emotional, and cognitive engagement. The analysis extends the discussion of existing frameworks of student engagement by incorporating insights specifically related to the use of ChatGPT in K12 and higher education (Fredricks et al., 2004; Zhao et al., 2023). Our findings from the meta-analyses revealed learning with ChatGPT fosters greater student behavioral, cognitive, and emotional engagement than learning without ChatGPT. The results from the systematic review suggest the major roles ChatGPT plays in promoting student engagement: PT, TA,

and CG. For example, student behavioral engagement is fostered by enhancing students' active participation and interaction with ChatGPT (PT), reflecting their involvement in learning activities, including technical suggestions and content generation (TA and CG). Cognitive engagement is fostered by getting instant feedback and clarification on complex topics (PT and TA) and generating videos and images (CG). Emotional engagement is fostered by seeking help without hesitation (PT). The results of this study were in line with those of previous systematic reviews and meta-analysis studies that discussed the effects of ChatGPT on student engagement (Li et al., 2024; Zhao et al., 2023).

Our systematic literature review revealed potential risks and results in student disengagement, such as over-reliance. The findings contribute to our understanding from both a quantitative and qualitative perspective. They encourage more teachers to integrate ChatGPT into learning and teaching. They can also serve as a foundation for developing instructional activities and evaluating student learning outcomes in ChatGPT-supported learning environments in future research.

Overall, the findings suggest that ChatGPT could be an effective learning tool for enhancing student engagement. The current educational dilemma is that the student adoption rate of ChatGPT in education is faster than that of teachers. These findings encourage teachers to use ChatGPT in teaching and learning to keep up with the student paces. Although this meta-analysis suggests ChatGPT has positive effects on student engagement, most of the selected articles are set in an intervention setting. The learning environments were under researcher control. These findings are driven by an experimental condition, which may not reflect the authentic learning environments. Therefore, we still need to be aware of the engagement effects, i.e., student disengagement brought by ChatGPT (Chiu, 2023, 2024).

### 5.2. Limitations and future research directions

This study has four limitations. First, to ensure the quality of the studies included in our analysis, we collected published empirical studies on applying ChatGPT interventions from journals. However, it is possible that papers from other databases, as well as grey literature (e.g., which include research or information that is not commercially published or peer-reviewed, such as conference proceedings, government reports, theses, dissertations, technical reports, and other unpublished works), or papers missed due to limitations in the search strings, may have been excluded from our search. In future research, the inclusion and exclusion criteria for selecting studies could be expanded to encompass other types of literature, allowing for the inclusion of relevant data from more diverse resources. Second, the number of included studies was relatively small for the meta-analysis due to the limited number of empirical studies that match the inclusion criteria. Therefore, future researchers are expected to collect more empirical studies on adopting different versions of ChatGPT or related large language models by focusing on its pedagogical applications. Third, this study compared the effects of student engagement in ChatGPT-incorporated learning activities to those of non-ChatGPT. However, some empirical studies have compared the effects of different disciplines or learning groups. For instance, Bilal Khallel (2024) compared the results of ChatGPT integration in independent, peer, and group learning, group. In future research, scholars can conduct meta-analyses with moderator analysis to compare the effects of individual and other learning groups. Fourth, this study analyzed the data from articles conducted at different educational levels due to the insufficient number of studies. The way K-12 students engage with ChatGPT should be different from university students. Therefore, we suggest future studies should focus on one educational level.

## CRediT authorship contribution statement

**Yuk Mui Elly Heung:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Thomas K.F. Chiu:** Writing – review & editing, Validation, Supervision, Methodology, Data curation, Conceptualization.

## Availability of supporting data

The datasets used for the current study are available from data repository.

## Funding

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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N/A.

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